

X3 Lab SIWES Roadmap for Project/Product Management

Advanced 6-Month Training Plan for a Technical Project Manager / Product Manager

Program Objective

This roadmap is designed to transform the intern from a student with technical ability into a **high-performing Technical Project Manager / Product Manager** who can:

- Lead cross-functional teams
 - Manage real product execution cycles
 - Build systems and workflows
 - Think strategically about products
 - Align business goals with engineering delivery
 - Own timelines, priorities, quality, and outcomes
 - Operate effectively in startup environments
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CORE TRAINING PILLARS

1. Product Management
 2. Technical Project Management
 3. Agile Delivery Systems
 4. Product Strategy
 5. Leadership & Communication
 6. Technical Understanding
 7. Data & Decision Making
 8. Innovation & Startup Thinking
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PERFORMANCE EXPECTATION AFTER 6 MONTHS

By completion, the intern should perform like:

Associate Product Manager
Technical Project Coordinator
Junior Delivery Manager
Startup Operations Lead (Entry Level)

MONTH 1 — PRODUCT OPERATIONS FOUNDATION

Goal: Understand How Products, Teams, and Startups Operate

WEEK 1: Orientation to X3 Lab Ecosystem

Learning Goals:

- Understand X3 Lab vision, mission, programs, products, partnerships
- Learn startup execution model
- Understand internal communication flow
- Learn role of PM in innovation organizations

Tasks:

- Attend leadership briefing
- Review existing projects
- Create org/process map
- Observe daily operations

Expected Outcomes:

Can explain how X3 Lab operates
Understands major business units
Knows where PM creates value

WEEK 2: Product Lifecycle Mastery

Learning Goals:

- Idea validation
- MVP concept
- Product-market fit basics
- Iteration cycles
- Growth stages

Tasks:

- Study one X3 Lab initiative from idea to delivery
- Document lifecycle stages
- Identify missing systems

Expected Outcomes:

Understands lifecycle deeply
Can map product stages professionally

WEEK 3: Project Management in Tech**Learning Goals:**

- Scope, timeline, cost, quality
- Dependencies
- Resource planning
- Risk concepts
- Delivery constraints

Tasks:

- Create sample project plan
- Build milestone chart
- Identify delivery risks

Expected Outcomes:

Can structure project plans
Understands real delivery pressure

WEEK 4: PM Tools Mastery**Tools:**

- Jira
- ClickUp
- Notion
- Slack
- Google Workspace

Tasks:

- Create sprint board
- Build dashboard
- Create documentation hub

Expected Outcomes:

Can manage modern workflows digitally

MONTH 2 — AGILE DELIVERY & EXECUTION

WEEK 5: Scrum Deep Dive**Learning Goals:**

- Roles in Scrum
- Sprint ceremonies
- Definition of Done
- Velocity
- Burndown logic

Tasks:

- Simulate sprint planning
- Participate in standups
- Document blockers

Outcomes:

Can run Scrum ceremonies

WEEK 6: Backlog Management**Learning Goals:**

- User stories
- Acceptance criteria
- Prioritization models (MoSCoW / RICE)
- Feature slicing

Tasks:

- Write 20 user stories
- Prioritize backlog for internal tool

Outcomes:

Can manage backlog professionally

WEEK 7: Execution Tracking**Learning Goals:**

- Delivery dashboards
- Timeline management
- Dependencies tracking
- Bottleneck detection

Tasks:

- Build tracker for live project
- Weekly status updates

Outcomes:

Can monitor execution in real time

WEEK 8: Risk & Escalation Management**Learning Goals:**

- Delivery risk identification
- Escalation frameworks
- Recovery planning

Tasks:

- Create risk register
- Solve delayed project simulation

Outcomes:

Can manage pressure professionally

MONTH 3 — TECHNICAL PRODUCT MANAGEMENT

WEEK 9: Systems Thinking for PMs

Learning Goals:

- Frontend/backend architecture basics
- APIs
- Databases
- Integrations
- Authentication systems

Tasks:

- Map one app architecture
- Explain user request to engineering terms

Outcomes:

Speaks technical language clearly

WEEK 10: PRD Writing Mastery

Learning Goals:

- Problem statements
- Goals
- Functional requirements
- Edge cases
- Success metrics

Tasks:

- Write PRD for X3 Lab internal product

Outcomes:

Writes strong product documents

WEEK 11: QA & Release Management

Learning Goals:

- Test cases
- Regression testing
- Bug severity levels
- Release checklist

Tasks:

- Conduct testing session
- Log issues

Outcomes:

Understands quality systems

WEEK 12: Engineering Collaboration

Learning Goals:

- Estimation methods
- Dev handoff
- Managing technical debt
- Sprint compromises

Tasks:

- Join developer review sessions
- Facilitate product-engineering alignment meeting

Outcomes:

Trusted by technical teams

MONTH 4 — PRODUCT STRATEGY & GROWTH

WEEK 13: Market Intelligence

Learning Goals:

- TAM / SAM / SOM basics
- Competitor research

- Positioning

Tasks:

- Analyze 3 competitors relevant to X3 Lab

Outcomes:

Thinks commercially

WEEK 14: User Research

Learning Goals:

- Interviews
- Surveys
- Pain-point extraction
- Personas

Tasks:

- Interview users/students/staff
- Build personas

Outcomes:

Builds user-first products

WEEK 15: Roadmapping

Learning Goals:

- Quarterly roadmap creation
- Capacity planning
- Tradeoffs

Tasks:

- Create 6-month roadmap for one product

Outcomes:

Can plan strategically

WEEK 16: Metrics & Growth

Learning Goals:

- Activation
- Retention
- Churn
- Funnel thinking
- North star metrics

Tasks:

- Build KPI dashboard

Outcomes:

Makes data-driven decisions

MONTH 5 — LEADERSHIP & EXECUTIVE READINESS

WEEK 17: Stakeholder Management

Learning Goals:

- Managing founders
- Managing dev teams
- Managing vendors
- Expectation control

Tasks:

- Present updates to leadership

Outcomes:

Executive communication ability

WEEK 18: Conflict Resolution

Learning Goals:

- Team tension management
- Scope disagreements
- Deadline conflicts

Tasks:

- Solve simulated project conflict cases

Outcomes:

Calm under pressure

WEEK 19: Meeting Leadership

Learning Goals:

- Running effective meetings
- Decision frameworks
- Follow-up systems

Tasks:

- Own weekly PM meeting

Outcomes:

Leads with authority

WEEK 20: Operational Excellence

Learning Goals:

- SOP creation
- Process optimization
- Automation thinking

Tasks:

- Improve one internal X3 Lab workflow

Outcomes:

Thinks like operator-leader

MONTH 6 — CAPSTONE EXECUTION

WEEK 21: Product Ownership Launch Planning

Tasks:

Choose one real project:

- Hackathon management platform
- Student onboarding system
- Training portal upgrade
- Dashboard for operations

Outcomes:

Owns real initiative

WEEK 22: Build & Coordinate

Tasks:

- Manage team tasks
- Follow up deadlines
- Remove blockers
- Quality checks

Outcomes:

Executes like PM

WEEK 23: Launch + Reporting

Tasks:

- Launch solution
- Gather feedback
- Analyze issues

Outcomes:

Understands go-live realities

WEEK 24: Executive Presentation

Final Deliverables:

- Product Requirement Document
- Roadmap
- Sprint tracker
- KPI report
- Lessons learned report
- Leadership presentation

Outcomes:

Ready for entry-level PM roles

WEEKLY OPERATING STRUCTURE

Day	Focus
Monday	Planning + Sprint priorities
Tuesday	Product documentation
Wednesday	Team coordination
Thursday	Strategy / Analysis
Friday	Reporting + Reflection

ADVANCED BOOKS / LEARNING RESOURCES

- Inspired – Marty Cagan
- Lean Startup – Eric Ries
- Scrum Guide

- Decode & Conquer (PM Interviews)
- Google PM Certificate
- Atlassian Agile Academy